# Unsupervised Encoder–Decoder Network Under Spatial and Spectral Guidance for Hyperspectral and Multispectral Image Fusion

Huajing Wu, Kefei Zhang, Suqin Wu, Shuangshuang Shi, Chaofa Bian, and Minghao Zhang

***Abstract*— Owing to the limitations of hyperspectral optical imaging, hyperspectral images (HSIs) have a dilemma between spectral and spatial resolutions. The hyperspectral and multispectral image (HSI–MSI) fusion, which combines a low-spatial-resolution HSI (LR-HSI) and a high-spatial-resolution multispectral image (HR-MSI), can generate a high-spatial-resolution HSI (HR-HSI). In existing methods for hyperspectral and multispectral fusion, correlation between spectral and spatial domains in HSIs is mostly neglected. To address this issue, an unsupervised encoder–decoder network under spatial and spectral guidance for the HSI–MSI fusion (uEDSSG) was proposed in this study. To learn more accurate abundances of an LR-HSI and an HR-MSI, multihierarchical encoders under spatial and spectral guidance were designed to extract multihierarchical fused features from the LR-HSI and HR-MSI with the guidance of the HR-MSI and LR-HSI, respectively. In the new method, deep coupling of the point spread function (PSF) or spectral response function (SRF) and edge of the HSIs was designed to maintain the spatial and spectral details of the HR-HSI; a spatial–spectral constraint was constructed to establish the relationship of the HSIs. Both visual and quantitative evaluation results of experiments based on both synthetic and real datasets showed that the proposed method outperformed seven common methods. The results suggest that the new method by maintaining the correlation between spectral and spatial domains can improve the result of HSI–MSI fusion.**

***Index Terms*— Deep coupling of point spread function (PSF)/spectral response function (SRF) and edge of hyperspectral images (HSIs), hyperspectral and multispectral image (HSI–MSI) fusion, multihierarchical encoder, and spatial and spectral guidance.**

## I. Introduction

HYPERSPECTRAL image (HSI) is a data cube containing multiple contiguous narrow-band images from the same scene [1]. The inherited high spectral resolution of HSI enables accurate identification of materials and therefore promotes various applications, such as classification [2], [3], detection of changes [4], [5], and detection of objects [6], [7]. Due to the limited incident energy of the Sun, the energy captured by each spectral band decreases with the increase in the number of spectral bands, resulting in a reduction in the spatial resolution of the images from these spectral bands. HSIs with a higher spectral resolution and multispectral images (MSIs) with a lower spectral resolution can be acquired with lower and higher spatial resolution, respectively. Therefore, there exists a dilemma between spectral and spatial resolutions that cannot be resolved by imaging sensors directly. To solve the dilemma, the fusion of a low-spatial-resolution HSI (LR-HSI) and a high-spatial-resolution MSI (HR-MSI), which is the so-called HSI–MSI fusion, can be used to obtain a high-spatial-resolution HSI (HR-HSI).

HSI–MSI fusion arises from the pan-sharpening, which fuses an LR-HSI with a high-spatial-resolution panchromatic image [8], [9], [10]. Compared with HSI–MSI fusion, pan-sharpening ignores spectral information, leading to spectral distortion of the result [11]. In contrast, the HSI–MSI fusion preserves more accurate spectral information with a high-spatial resolution. Traditional fusion methods can be divided into three categories [12]: the extensions of pan-sharpening methods, matrix factorization-based methods, and tensor factorization-based methods.

The extensions of pan-sharpening methods maintain spectral information based on pan-sharpening methods. The early methods extract the spatial components of an HR-MSI using principal component analysis (PCA) [13], [14], wavelet transformation [15], [16], and Laplacian pyramid [17], and then injected them into an LR-HSI to obtain an HR-HSI. However, the early methods only focus on spatial details, which often causes serious spectral distortion of the resultant HR-HSI. To reduce the spectral distortion, a Bayesian-based fusion method [18] used a stochastic mixing model of the underlying spectral content to improve spatial resolution. As an underdetermined inverse problem, HSI–MSI fusion requires a priori constraints. Total variational regularities [19] were added into the Bayesian model to resolve the illness of the HSI–MSI fusion.

The matrix factorization-based methods regard an HSI as a matrix of spatial dimension and spectral dimension and divide it into spectral basis and spatial coefficients by spectral

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unmixing [20]. The spectral basis is usually obtained from HSIs through a sparse dictionary [21], [22], [23] and low-rank learning [24], [25]. Robinson et al. [26] multiplied the spectral basis of an LR-HSI by the spatial coefficient matrix of an HR-MSI to obtain a potential HR-HSI. A coupling matrix decomposition scheme [27] was proposed to update the spectral bases and coefficients alternately. To preserve spatial details during matrix decomposition, low-rank a prior [24] was used to constrain the spatial coefficients of the potential HR-HSI. Dian et al. [28] proposed a nonlocal similarity to further ensure the accuracy of spatial coefficients in an HR-HSI.

The tensor factorization-based methods consider an HSI as a 3-D tensor containing spatial width, height, and spectrum [29]. Their procedure is as follows. First, an HSI can be decomposed into several tensor factors through tensor representation methods such as Tucker decomposition [30], Canonical Polyadic (CP) decomposition [31], [32], and singular value (t-SVD) decomposition [33]. Then, these factors can be solved from an LR-HSI and an HR-MSI. Finally, an HR-HSI can be constructed by these factors. To accurately characterize the spatial and spectral information of HSIs, some decompositions were proposed for the HSI–MSI fusion. Xu et al. [34] characterized unidirectional total variation on the Tucker decomposition factors. To accurately estimate the CP decomposition factor, a nonlocal CP decomposition for HSI–MSI fusion [31] assumed that the LR-HSI and HR-MSI have the same factor. On the basis of t-SVD, a new tensor multirank measure method [35] measured the structural complexity of HSIs. To further accurately represent HSIs, a nonlocal sparsity [28] and low-rank property [36], [37] fully made use of the inherent structures of 3-D HSIs.

In recent years, deep learning has been successfully applied to HSI–MSI fusion. Compared with traditional methods, the methods that are based on deep learning have shown superior performance due to their powerful feature-extraction ability [38], [39]. Xie at al. [40] designed an iterative algorithm to solve the HSI–MSI fusion model and learned the parameters of the model by constructing a deep network. Dian et al. [41] initialized an HR-HSI by solving a Sylvester equation and then learned the mapping between the initialized HR-HSI and target HR-HSI by a neural network. Huang et al. [42] learned the spatial mapping from an LR-MSI to an HR-MSI by a sparse denoising self-encoder and then improved the spatial resolution of each spectral band of an LR-HSI by the learned mapping. Li et al. [43] designed a dense-detail-insertion block for enhancing feature reuse and interaction. Wang et al. [44] proposed an unsupervised deep-learning-based method that considered interimage variability of data from heterogeneous sources and flexible image priors. To achieve more effective information, a multihierarchical cross-transformer was constructed for HSI–MSI fusion [45].

In HSIs, there is often the similarity between adjacent pixel spectra. Therefore, the spatial and spectral domains in HSIs are strongly relevant. Most of the aforementioned methods attempt to maintain the spatial and spectral details of the unknown HR-HSI but neglect the strong correlation between spectral and spatial domains in HSIs, resulting in poor fusion results.

To address the aforementioned problem, an unsupervised encoder–decoder network under spatial and spectral guidance for hyperspectral and MSI (HSI–MSI) fusion (uEDSSG) was proposed and its procedure is as follows. First, multihierarchical encoders under spatial and spectral guidance were designed to extract the abundances of an HR-MSI and an LR-HSI, respectively. Second, MSI and HSI decoders were used to construct an HR-MSI and an LR-HSI based on the abundances of the HR-MSI and LR-HSI, respectively. Then, deep coupling of the edge of HSIs and point spread function (PSF) or spectral response function (SRF) was constructed to maintain spatial and spectral details. Finally, a spatial–spectral constraint was designed to maintain the correlation between spectral and spatial domains the fused HR-HSI. Compared with the existing HSI–MSI approaches, the major innovative aspects of uEDSSG proposed by this study are highlighted as follows.

1) Both HSI–MSI fusion and PSF/SRF learning interact and promote each other.
2) To learn more accurate abundances, multihierarchical encoders under spatial and spectral guidance extract multihierarchical fused features from the LR-HSI and HR-MSI with the guidance of the HR-MSI and LR-HSI, respectively.
3) Deep coupling of the edge of HSIs and PSF/SRF, which establishes the relationship between the spatial and spectral edges of HSIs, is designed to maintain spatial and spectral details for a required HR-HSI.
4) A spatial–spectral constraint is constructed to explore the relationship of the HR-MSI and LR-HSI and maintain the correlation between spectral and spatial domains for a required HR-HSI.

The remainder of this article is organized as follows. Section II introduces problem formulation and an encoder–decoder network for HSI–MSI fusion. The proposed uEDSSG method is introduced in Section III, followed by Section IV for experimental results on both synthetic and real datasets. Conclusions are given in Section V.

## II. Related Work

### A. Problem Formulation

An HR-HSI, the target of HSI–MSI fusion, is expected to carry complete spatial and spectral information. Compared with an HR-HSI, an LR-HSI of the same scene has a lower spatial resolution. Every spectral band image of the LR-HSI can be regarded as spatial degradation from the corresponding spectral band image of the HR-HSI by a PSF and downsampling operation. Let $\boldsymbol{X} \in R^{H\times M\times N}$, $\boldsymbol{Y}_{\mathrm{H}} \in R^{H\times m\times n}$, and $\boldsymbol{Y}_{\mathrm{h}} \in R^{h\times M\times N}$ be the HR-HSI with $M \times N$ spatial resolution and $H$ spectral bands, the LR-HSI with $m \times n$ spatial resolution and $H$ spectral bands, and the HR-MSI with $M \times N$ spatial resolution and $h$ spectral bands, respectively. The relationship between $\boldsymbol{X}$ and $\boldsymbol{Y}_{\mathrm{H}}$ is often modeled as

$$\boldsymbol{Y}_{\mathrm{H}} = (\boldsymbol{X} * \boldsymbol{M})_{\downarrow} \tag{1}$$

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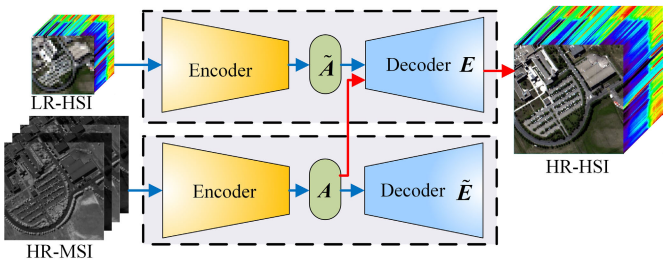



Fig. 1. Structure of the encoder–decoder network.

where $\boldsymbol{M} \in R^{Lm \times Ln}$ is the PSF, $*$ denotes the convolution operator, $Lm$ and $Ln$ are the spatial size of the convolution, and $(.)_{\downarrow}$ denotes the downsampling operation.

Compared with the HR-HSI, the HR-MSI of the same scene lacks complete spectral information. The spectral information of the HR-MSI can be regarded as a linear combination of the complete spectral information of the HR-HSI by an SRF. The relationship between $\boldsymbol{X}$ and $\boldsymbol{Y}_{\mathrm{h}}$ is often modeled as

$$\boldsymbol{Y}_{\mathrm{h}} = \boldsymbol{R}\boldsymbol{X} \tag{2}$$

where $\boldsymbol{R} \in R^{h \times H}$ is the SRF whose columns contain the spectral response.

### B. Encoder–Decoder Network

The encoder–decoder network is based on the hyperspectral linear unmixing model in which each pixel of an HSI is assumed to be a linear combination of a set of endmembers and the abundances of each endmember. The endmembers and abundances of the HR-HSI can be represented as $\boldsymbol{E} \in R^{H \times p}$ and $\boldsymbol{A} \in R^{p \times M \times N}$, respectively, where $p$ is the number of endmembers. According to the linear spectral unmixing model, the target HR-HSI can be expressed as

$$\boldsymbol{X} = \boldsymbol{E}\boldsymbol{A}. \tag{3}$$

The above equation describes the degree of mixing for each pixel in the HR-HSI $\boldsymbol{X}$. An LR-HSI can be decomposed to endmembers $\boldsymbol{E}$ and the abundances $\tilde{\boldsymbol{A}} \in R^{p \times m \times n}$, and an HR-MSI can be decomposed to endmembers $\tilde{\boldsymbol{E}} \in R^{h \times p}$ and the abundances $\boldsymbol{A}$. According to (1) and (2), the LR-HSI and HR-MSI can be expressed as

$$\begin{cases} \boldsymbol{Y}_{\mathrm{H}} = \boldsymbol{E}(\boldsymbol{A} * \boldsymbol{M})_{\downarrow} = \boldsymbol{E}\tilde{\boldsymbol{A}} \\ \boldsymbol{Y}_{\mathrm{h}} = \boldsymbol{R}\boldsymbol{E}\boldsymbol{A} = \tilde{\boldsymbol{E}}\boldsymbol{A}. \end{cases} \tag{4}$$

The encoder–decoder network is based on (3) and (4). As shown in Fig. 1, the two encoders and two decoders extract abundances and endmembers from an LR-HSI and an HR-MSI. In the encoder–decoder network, an HR-HSI is obtained by the abundances $\boldsymbol{A}$ from the HR-MSI and endmembers $\boldsymbol{E}$ from the LR-HSI. Therefore, the accurate decomposition is the key to effective fusion. At present, there are two methods that can be used to improve the decomposition. One is through designing an appropriate network structure, and the other is through seeking the mutual constraining relationship between the LR-HSI and HR-MSI.

Various encoder–decoder network structures have been designed by scholars. Uezato et al. [46] extracted semantic features from the guidance image by the U-net. Three coupled autoencoder networks composed of convolution kernel of size $1 \times 1$ and LReLU layers repeatedly [47] were designed to encode the dimension reduction of the spectral direction of HSIs. In addition, more precise constraints of an encoder have also been proposed. An attention mechanism [48] was used to constrain the space and spectrum of the LR-HSI and HR-MSI, respectively, during the process of encoding. An unsupervised Sparse Dirichlet Net used angle similarity to make the encoders of the LR-HSI and HR-MSI mutually constrained [38].

Inspired by the aforementioned methods, in the proposed uEDSSG method, the encoder–decoder network and a precise constraint were designed to construct the spatial and spectral details of an HR-HSI and to maintain the correlation between spectral and spatial domains in the HR-HSI, respectively.

## III. Method

The uEDSSG method proposed by this study is composed of two encoder–decoder subnetworks. First, the multihierarchical encoders under spatial and spectral guidance were for extracting abundances from an LR-HSI and an HR-MSI with the guidance/aid of the spatial information of the HR-MSI and the spectral information of the LR-HSI, respectively. They enable the HSIs to be decomposed more accurately. Second, for maintaining the spatial–spectral structure, deep coupling of edge and PSF/SRF was designed to explore the relationship of the spatial and spectral edge among the LR-HSI, HR-MSI, and HR-HSI. Then, a spatial–spectral constraint was designed to use the spatial–spectral consistency among the LR-HSI, HR-MSI, LR-MSI, and HR-HSI for constraining the entire network architecture. Finally, the loss function was established for the reconstruction of the fused HR-HSI.

### A. Network Structure

The overall network structure of uEDSSG composed of two encoder–decoder subnetworks includes a multihierarchical encoder under spectral guidance, a multihierarchical encoder under spatial guidance, an MSI decoder, and an HSI decoder, as shown in Fig. 2. The multihierarchical encoders under spatial and spectral guidance were designed to extract abundances from an HR-MSI and an LR-HSI, respectively. To supplement the spatial information of the LR-HSI and the spectral information of the HR-MSI, spatial information from the HR-MSI and spectral information from the LR-HSI were used to aid the extraction of abundances from the LR-HSI and HR-MSI, respectively. The two decoders (the MSI decoder and the HSI decoder) with spectral and spatial information were used to reconstruct the HR-MSI and LR-HSI. A fused HR-HSI was obtained through the MSI encoder and HSI decoder, which output spatial feature and spectral information from the HR-MSI and the LR-HSI, respectively. In addition, two convolution layers with kernel size $r \times r$ and $1 \times 1$ followed by a normalization layer were used to simulate the PSF and SRF, respectively.

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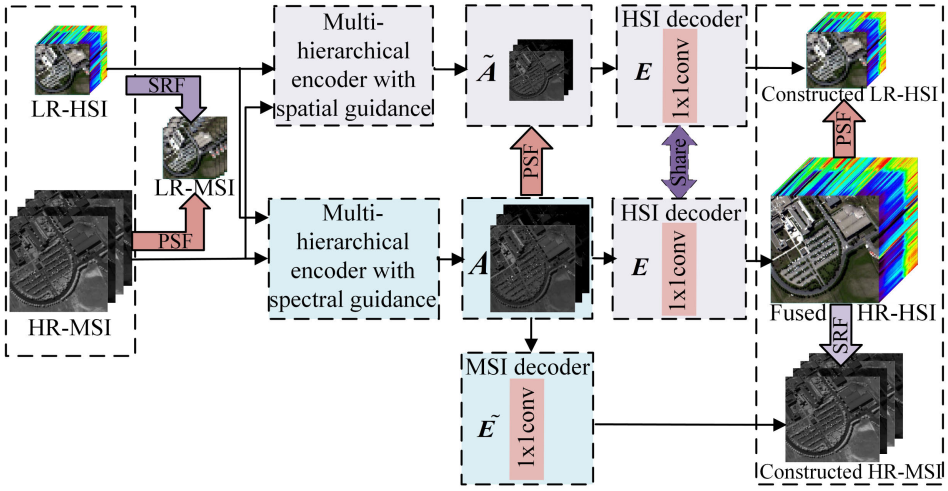



Fig. 2. Overall network structure of uEDSSG.

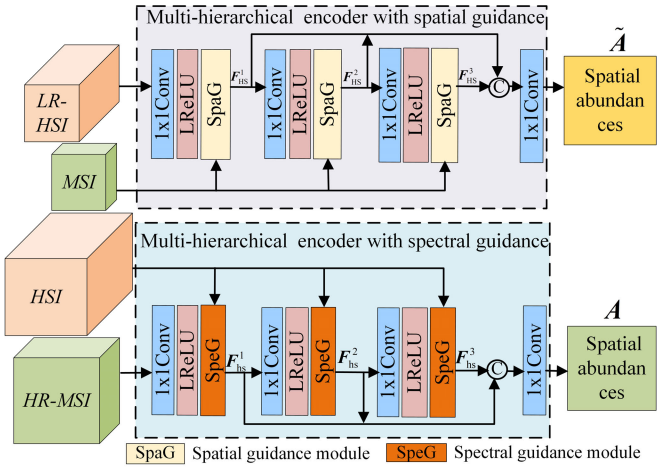



Fig. 3. Multihierarchical encoders under spatial and spectral guidance.

### B. Multihierarchical Encoder Under Spatial Guidance

Fig. 3 shows the multihierarchical encoders under spatial and spectral guidance designed for a supplement to the spatial and spectral information in the extraction of the abundances of an LR-HSI and an HR-MSI. In the multihierarchical encoder under spatial guidance, spatial information from an HR-MSI is injected at multiple levels as the spatial guidance for obtaining more accurate abundances of the LR-HSI.

To extract abundances from an LR-HSI more accurately, in the multihierarchical encoder under spatial guidance, the LR-MSI is employed in multihierarchical features. First, the HR-MSI is transformed into the same spatial size of the LR-HSI by spatial downsampling, and a transformed MSI $\boldsymbol{Y}'_{\mathrm{h}} \in R^{h\times m\times n}$ is obtained. Next, a 2-D convolution layer with a kernel size of $1 \times 1$ and an LReLU layer obtained first-hierarchical features $F^1_{\mathrm{H}}$ from the LR-HSI. Then, $\boldsymbol{Y}'_{\mathrm{h}} \in R^{h\times m\times n}$ is fused with $F^1_{\mathrm{H}}$ by a spatial guidance module, and the first-hierarchical fused features $\boldsymbol{F}^1_{\mathrm{HS}} \in R^{C_1\times m\times n}$ are obtained. Lastly, the second and third steps are repeated, and multihierarchical fused features $\boldsymbol{F}_{\mathrm{HS}}\{\boldsymbol{F}^1_{\mathrm{HS}}, \boldsymbol{F}^2_{\mathrm{HS}}, \boldsymbol{F}^3_{\mathrm{HS}}\}$ are obtained. These features can be defined as

$$\begin{cases} \boldsymbol{F}^1_{\mathrm{H}} = \mathrm{LRelu}(\mathrm{conv}(\boldsymbol{Y}_{\mathrm{H}})) \\ \boldsymbol{F}^i_{\mathrm{H}} = \mathrm{LRelu}\left(\mathrm{conv}\left(\boldsymbol{F}^{i-1}_{\mathrm{HS}}\right)\right) \quad \text{when } i > 1 \\ \boldsymbol{F}^i_{\mathrm{HS}} = \mathrm{SpatialGuidance}\left(\boldsymbol{F}^i_{\mathrm{H}}, \boldsymbol{Y}'_{\mathrm{h}}\right). \end{cases} \tag{5}$$

Finally, the multihierarchical fused features FH were concatenated and then transformed into by a 2-D convolution layer with a kernel size of $1 \times 1$. It can be defined as

$$\tilde{\boldsymbol{A}} = f_{\mathrm{en}}\left(\boldsymbol{Y}_{\mathrm{H}}, \boldsymbol{Y}'_{\mathrm{h}}\right) = \mathrm{conv}\left(\mathrm{concat}\left(\boldsymbol{F}^1_{\mathrm{HS}}, \boldsymbol{F}^2_{\mathrm{HS}}, \boldsymbol{F}^3_{\mathrm{HS}}\right)\right). \tag{6}$$

*1) Spatial Guidance Module:* To inject spatial details into the original features, a spatial guidance module was built (see Fig. 4) for its architecture. More specifications are as follows. First, $\boldsymbol{Y}'_{\mathrm{h}} \in R^{h\times m\times n}$ is followed by a 2-D convolution layer to expand the channel dimension from $h$ to $Ci$, and the spatial features $\boldsymbol{F}'_{\mathrm{h}} \in R^{C_i\times m\times n}$ are obtained. Then, a dimensionality reduction operation flattens on the input feature maps $\boldsymbol{F}^i_{\mathrm{H}} \in R^{C_i\times m\times n}$ and $\boldsymbol{F}'_{\mathrm{h}} \in R^{C_i\times m\times n}$ to obtain two $Ci \times D$ $(D = m \times n)$ sequences $\left(\boldsymbol{F}^i_{\mathrm{H}}\right)_{\mathrm{sq}}$ and $\left(\boldsymbol{F}'_{\mathrm{h}}\right)_{\mathrm{sq}}$. Each vector of two sequences produces queries, keys, and values of the attention operation, as shown in Fig. 4. We used $\boldsymbol{Q}^j_{\mathrm{H}}$, $\boldsymbol{K}^j_{\mathrm{H}}$, and $\boldsymbol{V}^j_{\mathrm{H}}$ to represent the $j$th vector of sequence $\left(\boldsymbol{F}^i_{\mathrm{H}}\right)_{\mathrm{sq}}$, $\boldsymbol{Q}^j_{\mathrm{h}}$, $\boldsymbol{K}^j_{\mathrm{h}}$ and $\boldsymbol{V}^j_{\mathrm{h}}$ to represent the $j$th vector of sequence $\left(\boldsymbol{F}'_{\mathrm{h}}\right)_{\mathrm{sq}}$. The spatial guidance can be expressed as

$$\begin{cases} \boldsymbol{K}^j_m = \mathrm{concat}\left(\boldsymbol{K}^j_{\mathrm{H}}, \boldsymbol{K}^j_{\mathrm{h}}\right) \ \boldsymbol{Q}^i_m = \mathrm{concat}\left(\boldsymbol{Q}^j_{\mathrm{H}}, \boldsymbol{Q}^j_{\mathrm{h}}\right) \\ \boldsymbol{Attention}^j_{\mathrm{H}} = \mathrm{soft\,max}\left(\boldsymbol{Q}^j_m\left(\boldsymbol{K}^j_m\right)^T \Big/ \sqrt{d}\right)\boldsymbol{V}^j_{\mathrm{H}} \\ \boldsymbol{Attention}^j_{\mathrm{h}} = \mathrm{soft\,max}\left(\boldsymbol{Q}^j_m\left(\boldsymbol{K}^j_m\right)^T \Big/ \sqrt{d}\right)\boldsymbol{V}^j_{\mathrm{h}} \\ \boldsymbol{F}^i_{\mathrm{HS}} = \mathrm{conv}(\mathrm{concat}(\boldsymbol{Attention}_{\mathrm{H}}, \boldsymbol{Attention}_{\mathrm{h}})) \end{cases} \tag{7}$$

where $d$ is the dimension of the key, $\boldsymbol{Attention}_{\mathrm{H}}$ and $\boldsymbol{Attention}_{\mathrm{h}}$ are the attention maps of multihierarchical fused features and the spatial features from the LR-HSI and HR-MSI, respectively, and the multihierarchical fused features $\boldsymbol{F}^i_{\mathrm{HS}}$ are concatenated and processed by a linear projection.

Due to the lack of spatial information in an LR-HSI, the multihierarchical encoder under spatial guidance imports

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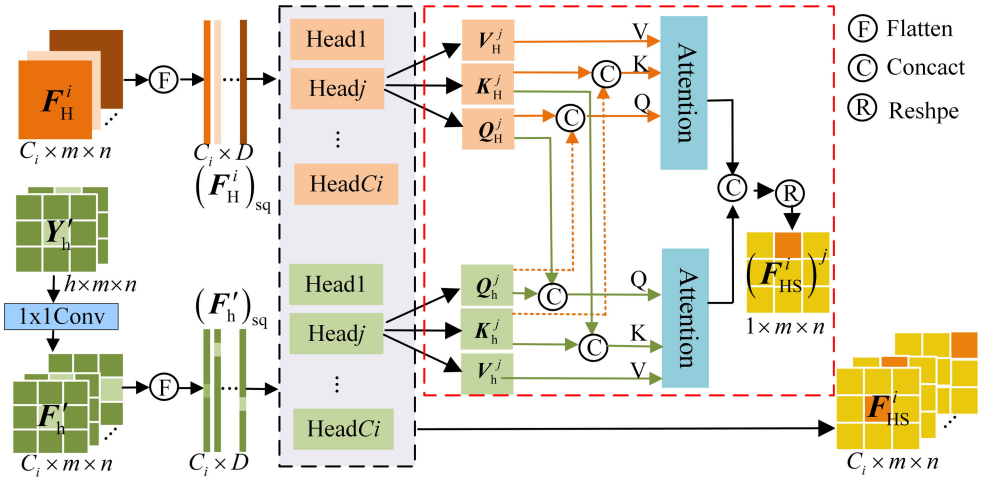



Fig. 4. Spatial guidance module.

spatial information from an HR-MSI and guides the extraction of the abundances in the LR-HSI. In the encoder, a spatial guidance module is built into inject spatial information of the HR-MSI into multihierarchical features from the LR-HSI, which achieves deep spatial fusion between the HR-MSI and LR-HSI.

### C. Multihierarchical Encoder Under Spectral Guidance

Similar to the lack of spatial information in an LR-HSI, an HR-MSI lacks spectral information, which hinders the extraction of abundances from the HR-MSI. To obtain more accurate abundances, the multihierarchical encoder under spectral guidance designed is as shown in Fig. 3. In the encoder, spectral information from an LR-HSI is fused with multihierarchical features from an HR-MSI through a spectral guidance module. The multihierarchical encoder under spectral guidance employs the LR-HSI in the multihierarchical features. First, the LR-HSI is transformed into the same spatial size of the HR-MSI by pixel shuffle, and a transformed HSI $\boldsymbol{Y}'_{\mathrm{H}} \in R^{H\times M\times N}$ is obtained. Next, a 2-D convolution layer with a kernel size of $1\times 1$ and an LReLU layer obtains first-hierarchical features $\boldsymbol{F}^1_{\mathrm{h}}$ from the HR-MSI. Then, $\boldsymbol{Y}'_{\mathrm{H}}$ is fused with $\boldsymbol{F}^1_{\mathrm{h}}$ by a spatial guidance module, and first-hierarchical fused features $\boldsymbol{F}^1_{\mathrm{hs}} \in R^{L_1\times M\times N}$ are obtained. Lastly, the second and third steps are repeated, and multihierarchical fused features $\boldsymbol{F}_{\mathrm{hs}}\{\boldsymbol{F}^1_{\mathrm{hs}}, \boldsymbol{F}^2_{\mathrm{hs}}, \boldsymbol{F}^3_{\mathrm{hs}}\}$ are obtained. These features can be defined as

$$\begin{cases} \boldsymbol{F}^1_{\mathrm{h}} = \mathrm{LRelu}(\mathrm{conv}(\boldsymbol{Y}_{\mathrm{h}})) \\ \boldsymbol{F}^i_{\mathrm{h}} = \mathrm{LRelu}\left(\mathrm{conv}\left(\boldsymbol{F}^{i-1}_{\mathrm{hs}}\right)\right) \quad \text{when } i > 1 \\ \boldsymbol{F}^i_{\mathrm{hs}} = \mathrm{SpectralGuidance}\left(\boldsymbol{F}^i_{\mathrm{h}}, \boldsymbol{Y}'_{\mathrm{H}}\right). \end{cases} \tag{8}$$

Finally, the multihierarchical fused features $\boldsymbol{F}_{\mathrm{hs}}$ are concatenated and then transformed into $\boldsymbol{A}$ by a 2-D convolution layer with a kernel size of $1\times 1$. It can be defined as

$$\boldsymbol{A} = g_{\mathrm{en}}\left(\boldsymbol{Y}_{\mathrm{h}}, \boldsymbol{Y}'_{\mathrm{H}}\right) = \mathrm{conv}\left(\mathrm{concat}\left(\boldsymbol{F}^1_{\mathrm{hs}}, \boldsymbol{F}^2_{\mathrm{hs}}, \boldsymbol{F}^3_{\mathrm{hs}}\right)\right). \tag{9}$$

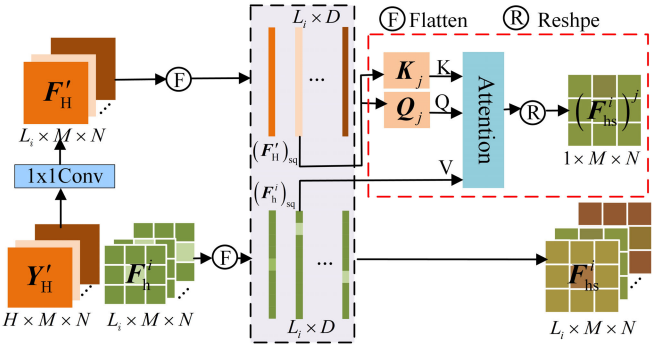



Fig. 5. Spectral guidance module.

*1) Spectral Guidance Module:* For injecting spectral information into the original features, the spectral guidance module proposed by this study was built and its architecture is illustrated in Fig. 5. Its procedure is described below. First, $\boldsymbol{Y}'_{\mathrm{H}}$ is followed by a 2-D convolution layer to expand the channel dimension from H to Li, and the spectral features $\boldsymbol{F}'_{\mathrm{H}} \in R^{L_i\times M\times N}$ are obtained. Then, a flattening operation is conducted on the input feature maps $\boldsymbol{F}^i_{\mathrm{h}} \in R^{L_i\times M\times N}$ and $\boldsymbol{F}'_{\mathrm{H}} \in R^{L_i\times M\times N}$ to obtain two $Li\times D$ $(D = M\times N)$ sequences $\left(\boldsymbol{F}^i_{\mathrm{h}}\right)_{\mathrm{sq}}$ and $\left(\boldsymbol{F}'_{\mathrm{H}}\right)_{\mathrm{sq}}$. The spectral guidance module can be expressed as

$$\begin{cases} \boldsymbol{K} = \boldsymbol{Q} = \left(\boldsymbol{F}_{\mathrm{H}}{}'\right)_{\mathrm{sq}}\boldsymbol{V} = \left(\boldsymbol{F}^i_{\mathrm{h}}\right)_{\mathrm{sq}} \\ \boldsymbol{F}^i_{\mathrm{hs}} = \boldsymbol{Attention} = \mathrm{soft\,max}\left(\boldsymbol{Q}(\boldsymbol{K})^T \Big/ \sqrt{d}\right)\boldsymbol{V} \end{cases} \tag{10}$$

where $d$ represents the dimension of the key.

The multihierarchical encoder under spectral guidance fuses spectral features from an HR-MSI with multihierarchical features by the spectral guidance module, which achieves deep spectral fusion between the HR-MSI and LR-HSI. The injection of the spectral information of the LR-HSI improves the abundances extraction of the HR-MSI.

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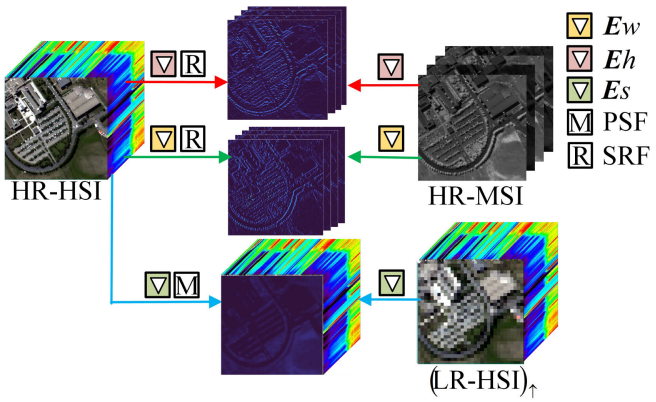



Fig. 6. Coupling of PSF/SRF and edge of HSIs.

### D. Construction of HSIs

For reconstructing HSIs, an HSI and an MSI decoder, which are expected to carry the spectral information of endmembers of an LR-HSI and an HR-MSI, respectively, were designed for reconstructing an LR-HSI and an HR-MSI with abundances, respectively. It is noted that the spectral information from the HR-MSI and LR-HSI is hidden in the MSI and HSI decoder parameters, respectively. The two decoders consist of a convolution layer with kernel size $1 \times 1$ as shown in Fig. 2. The two encoder functions $f_{\text{de}}()$ and $g_{\text{de}}()$ can be expressed as

$$\begin{cases} f_{\text{de}}(\tilde{\boldsymbol{A}}) = \boldsymbol{Y}_{\text{H}} \\ g_{\text{de}}(\boldsymbol{A}) = \boldsymbol{Y}_{\text{h}}. \end{cases} \tag{11}$$

Compared with $\tilde{\boldsymbol{A}}$ of the LR-HSI, $\boldsymbol{A}$ of the HR-MSI contains more spatial information. Compared with the MSI decoder, the HSI decoder has more abundant spectral information. Therefore, the HR-HSI is the output of the HSI encoder fed with $\boldsymbol{A}$. The fused HR-HSI can be expressed as

$$\boldsymbol{X} = f_{\text{de}}(\boldsymbol{A}). \tag{12}$$

### E. Coupling of PSF/SRF and Edge of HSIs

The spatial and spectral edges introduced into the network are for maintaining spatial and spectral details. The relationship between the spatial edges of an HR-MSI and an HR-HSI is explored, which deeply couples the SRF and the spatial edge of the two HSIs. The relationship between the spectral edges of the LR-HSI and HR-HSI is also explored, which deep couples the PSF and the spectral edge of the two HSIs, as shown in Fig. 6. The deep coupling of the PSF/SRF and the edge of HSIs can maintain the spatial and spectral structure of the HR-HSI.

The spatial and spectral edges of an image cube $\boldsymbol{Z}$ can be defined as

$$\begin{cases} \boldsymbol{E}w_{k,i,j,}(\boldsymbol{Z}) = \boldsymbol{Z}_{k,i+1,j} - \boldsymbol{Z}_{k,i,j} \\ \boldsymbol{E}h_{k,i,j}(\boldsymbol{Z}) = \boldsymbol{Z}_{k,i,j+1} - \boldsymbol{Z}_{k,i,j} \\ \boldsymbol{E}s_{k,i,j}(\boldsymbol{Z}) = \boldsymbol{Z}_{k+1,i,j} - \boldsymbol{Z}_{k,i,j} \end{cases} \tag{13}$$

where $i$ and $j$ are the indices of the spatial positions along the width and height directions, respectively, $k$ is the index of the spectral position, $\boldsymbol{E}w_{k,i,j}(\boldsymbol{Z})$ and $\boldsymbol{E}h_{k,i,j}(\boldsymbol{Z})$ are the edges of $\boldsymbol{Z}_{k,i,j}$ along the spatial width and height directions, respectively, and $\boldsymbol{E}s_{k,i,j}(\boldsymbol{Z})$ is the spectral edge of $\boldsymbol{Z}_{k,i,j}$. The edge map can be obtained as Fig. 6.

The fused HR-HSI is expected to have complete spatial information. An HR-MSI with complete spatial information can provide spatial edges along the width direction to maintain the spatial structure of the HR-HSI. According to (2), the relationship of the spatial edges along the width between the HR-MSI and HR-HSI is expressed as

$$\begin{aligned} \boldsymbol{E}w_{k,i,j}(\boldsymbol{Y}_{\text{h}}) &= (\boldsymbol{RX})_{k,i+1,j} - (\boldsymbol{RX})_{k,i,j} \\ &= \sum_{h=1}^{H} \left(\boldsymbol{R}_{k,h}\boldsymbol{X}_{h,i+1,j}\right) - \sum_{h=1}^{H} \left(\boldsymbol{R}_{k,h}\boldsymbol{X}_{h,i,j}\right) \\ &= \sum_{h=1}^{H} \left(\boldsymbol{R}_{k,h}\boldsymbol{X}_{h,i+1,j} - \boldsymbol{R}_{k,h}\boldsymbol{X}_{h,i,j}\right) \\ &= \sum_{h=1}^{H} \boldsymbol{R}_{k,h}\left(\boldsymbol{X}_{h,i+1,j} - \boldsymbol{X}_{h,i,j}\right) \\ &= \sum_{h=1}^{H} \boldsymbol{R}_{k,h}\left(\boldsymbol{E}w_{h,i,j}(\boldsymbol{X})\right) \\ &= \boldsymbol{RE}w_{k,i,j}(\boldsymbol{X}). \end{aligned} \tag{14}$$

Similar to (14), the relationship between the spatial edges of the HR-MSI and HR-HSI along height can be expressed as $\boldsymbol{E}h_{k,i,j}(\boldsymbol{Y}_{\text{h}}) = \boldsymbol{RE}h_{k,i,j}(\boldsymbol{X})$.

Similarly, the fused HR-HSI is also expected to have complete spectral information as well. An LR-HSI with complete spectral information can provide spectral edges for maintaining the spectral structure of the HR-HSI. The LR-HSI upsampled to the same size as the HR-HSI by linear interpolation can be denoted as $(\boldsymbol{Y}_{\text{H}})_{\uparrow}$. According to (1), the relationship between the spectral edges of the LR-HSI and HR-HSI is expressed as

$$\begin{aligned} \boldsymbol{E}s_{k,i,j}\left((\boldsymbol{Y}_{\text{H}})_{\uparrow}\right) &= \boldsymbol{E}s_{k,i,j}(\boldsymbol{X} * \boldsymbol{M}) = \boldsymbol{E}s_{k,i,j}(\boldsymbol{XC}_{\boldsymbol{M}}) \\ &= (\boldsymbol{XC}_{\boldsymbol{M}})_{k+1,i,j} - (\boldsymbol{XC}_{\boldsymbol{M}})_{k,i,j} \\ &= \sum_{d=1}^{M\times N} \left(\boldsymbol{x}_{k+1,d}\boldsymbol{C}_{\boldsymbol{M}d,i\times j}\right) - \sum_{d=1}^{M\times N} \left(\boldsymbol{x}_{k,d}\boldsymbol{C}_{\boldsymbol{M}d,i\times j}\right) \\ &= \sum_{d=1}^{M\times N} \left(\boldsymbol{x}_{k+1,d}\boldsymbol{C}_{\boldsymbol{M}d,i\times j} - \boldsymbol{x}_{k,d}\boldsymbol{C}_{\boldsymbol{M}d,i\times j}\right) \\ &= \sum_{d=1}^{M\times N} \left(\left(\boldsymbol{x}_{k+1,d} - \boldsymbol{x}_{k,d}\right)\boldsymbol{C}_{\boldsymbol{M}d,i\times j}\right) \\ &= \sum_{d=1}^{M\times N} \left(\left(\boldsymbol{E}s_{k,d}(\mathcal{X})\right)\boldsymbol{C}_{\boldsymbol{M}d,i\times j}\right) \\ &= \left(\boldsymbol{E}s_{k,i,j}(\boldsymbol{X})\right)\boldsymbol{C}_{\boldsymbol{M}} = \boldsymbol{E}s_{k,i,j}(\boldsymbol{X}) * \boldsymbol{M} \end{aligned} \tag{15}$$

where $\boldsymbol{C}_{\boldsymbol{M}}$ is the spatial degradation operator, which is often assumed to be composed of a cyclic convolution operator from $\boldsymbol{M}$ and $\boldsymbol{x} \in R^{H\times MN}$ is the expanded format of $\boldsymbol{X} \in R^{H\times M\times N}$. It should be noted that $i \times j$ and $j \times i$ denote two different spatial locations in the HSIs.

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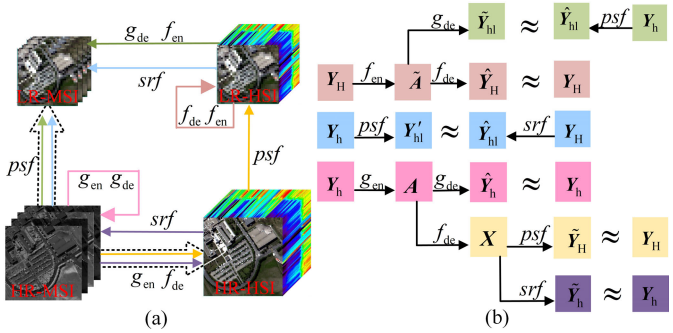



Fig. 7. (a) Spatial–spectral constraint of the network consisting four HSIs and (b) corresponding relationship among the four HSIs.

The coupling of the PSF and spatial edge of the HSIs is for maintaining the spatial details of the fused HR-HSI by establishing a relationship between the spatial edges of the HR-MSI and HR-HSI. Similarly, the coupling of the SRF and the spectral edge of the HSIs is for maintaining the spectral details of the HR-HSI by establishing a relationship between the spectral edges of the LR-HSI and HR-HSI. As a result, the resultant HR-HSI have the same spatial and spectral details as the HR-MSI and LR-HSI, respectively.

### F. Spatial–Spectral Constraint

The spatial–spectral constraint of the network among the four HSIs: an LR-HSI, an HR-MSI, an HR-HSI, and an LR-MSI is shown in Fig. 7(a), in which each arrow denotes a relationship expressed by the equation listed in Fig. 7(b) in the same color. For example, the yellow arrow denotes that the HR-HSI can be obtained through the MSI encoder and HSI decoder with the HR-MSI, then the HR-HSI followed the PSF and SRF network are equal to the LR-HSI and HR-MSI, respectively. The spatial–spectral constraint of the network can be expressed as

$$\begin{aligned}\mathcal{L}_{\text{spaspe}} = &\left\|\hat{\boldsymbol{Y}}_{\text{H}} - \boldsymbol{Y}_{\text{H}}\right\| + \alpha\left\|\tilde{\boldsymbol{Y}}_{\text{hl}} - \hat{\boldsymbol{Y}}_{\text{hl}}\right\| + \beta\left\|\boldsymbol{Y}'_{\text{hl}} - \hat{\boldsymbol{Y}}_{\text{hl}}\right\| \\ &+ \gamma\left\|\hat{\boldsymbol{Y}}_{\text{h}} - \boldsymbol{Y}_{\text{h}}\right\| + \eta\left\|\tilde{\boldsymbol{Y}}_{\text{H}} - \boldsymbol{Y}_{\text{H}}\right\| + \lambda\left\|\tilde{\boldsymbol{Y}}_{\text{h}} - \boldsymbol{Y}_{\text{h}}\right\|.\end{aligned} \tag{16}$$

### G. Joint Loss Function

For constructing a joint loss function, several constraints were added to the network, and the deep coupling of the PSF/SRF and edge of HSIs can be expressed as

$$\begin{aligned}\mathcal{L}_{\nabla} = &\left(\boldsymbol{E}w_{k,i,j}(\boldsymbol{Y}_{\text{h}}) - \boldsymbol{R}\boldsymbol{E}w_{k,i,j}(\boldsymbol{X})\right) \\ &+ \left(\boldsymbol{E}h_{k,i,j}(\boldsymbol{Y}_{\text{h}}) - \boldsymbol{R}\boldsymbol{E}h_{k,i,j}(\boldsymbol{X})\right) \\ &+ \left(\boldsymbol{E}s_{k,i,j}\left((\boldsymbol{Y}_{\text{H}})_{\uparrow}\right) - \boldsymbol{E}s_{k,i,j}(\boldsymbol{X}) * \boldsymbol{M}\right).\end{aligned} \tag{17}$$

To satisfy the sum-to-one and nonnegative properties [47] of the abundances, the sum of the abundances in the channel dimension is constrained to meet the sum-to-one requirement, which can be denoted by

$$\mathcal{L}_{\text{sum2one}} = \left\|1 - \sum_{i=1}^{p}\tilde{A}\right\|_1 + \left\|1 - \sum_{i=1}^{p}A\right\|_1, \quad \tilde{A}\succeq 0\ A\succeq 0 \tag{18}$$

where $\succeq$ marks element-wise inequality, which guarantees the nonnegative constraint.

Since each pixel of an HSI is composed of a small number of pure spectral bases, the abundances and are likely to be sparse. To ensure the sparsity of the abundances, the KL divergence is used to ensure that most of the elements in the abundances are close to a small number. The sparsity of the abundances can be expressed as

$$\mathcal{L}_{\text{sparse}} = KL\left(\tilde{\boldsymbol{A}}\right) + KL(\boldsymbol{A}). \tag{19}$$

Finally, the joint loss function can be formed by a combination of the following four parts: the spatial–spectral constraint, coupling of PSF/SRF and edge of HSIs, sum-to-one property, and sparsity of the abundances. Then, the fusion problem can be transformed into the optimization problem with the loss function, which can be defined as

$$\mathcal{L} = \mathcal{L}_{\text{spaspe}} + \mathcal{L}_{\nabla} + \mu\mathcal{L}_{\text{sum2one}} + \nu\mathcal{L}_{\text{sparse}}. \tag{20}$$

## IV. Experimental Results and Analysis

In this section, the results of experiments conducted for evaluating the performance of the proposed uEDSSG are presented. First, the activation function and the optimal number of endmembers are discussed and determined based on related experimental results and then ablation test results are analyzed. Second, the results of the PSF with different spatially downsampled kernels and the SRF are analyzed based on synthetic datasets. Finally, the fused images resulting from different methods and either synthetic and real datasets were compared using both visual and quantitative measures for their performance assessment.

### A. Implementation Details

In the experiment, the dimension of $\boldsymbol{F}_{\text{HS}}\{\boldsymbol{F}^1_{\text{HS}}, \boldsymbol{F}^2_{\text{HS}}, \boldsymbol{F}^3_{\text{HS}}\}$ was set to 64, 128, 256 and that of was set to {64, 128, 256}. The proposed network was implemented using the PyTorch framework, and the Adam optimizer was used to train the network for 10 000 epochs. The initial learning rate and weight decay were set to 0.005 and 0.0005, respectively; The number of input images was one. Therefore, each epoch equates to one iteration and the batch size is also one. The training environment consisted of an Intel i7-6850K CPU, 128-GB RAM, and NVIDIA TITAN Xp 12G GPU.

### B. Dataset

For experiments, three synthetic datasets were used as the testing data, and one real dataset was used as the reference for the evaluation of the test results from this study. The three synthetic datasets were from the Pavia University, Indian Pines, and Washington DC, and the real dataset was from Houston University. More details for each of the datasets are introduced below.

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*1) Three Synthetic Datasets:* The first dataset (i.e., Pavia University) was acquired from the reflective optics system imaging spectrometer ROSIS sensor during a flight campaign at Pavia University, northern Italy. The image is of size $610 \times 340 \times 115$ with a 1.3-m spatial resolution and the spectral coverage ranging from 0.43 to 0.86 $\mu$m. Due to the effects of noise and water vapor absorption, 12 bands were removed. An area covering $256 \times 256$ pixels in the lower-left corner of the image and containing 103 bands was selected for the experiment. The second dataset (i.e., Indian Pines) was acquired by the A VIRIS sensor. The images are of size $145 \times 145 \times 224$ with a 20-m spatial resolution and the spectral coverage ranging from 0.4 to 2.5 $\mu$m. After 33 noisy bands were removed, a $144 \times 144$-pixel area with 191 bands was remained and used as the experimental dataset. The third dataset (i.e., Washington DC) was taken by the HYDICE sensor. It is of size $1280 \times 307 \times 210$ with a 2.5-m spatial resolution and the spectral coverage ranging from 0.4 to 2.5 $\mu$m. After ten noisy bands were removed, an area covering $256 \times 256$ pixels in the lower-left corner of the image containing 200 bands was selected for the experiment.

The aforementioned three datasets plus a simulated PSF and SRF were used to generate two observed images: LR-HSIs and HR-MSIs, based on which the three new synthetic datasets were formed. More specifically, in our experiment, a Gaussian blur of $15 \times 15$ pixels with a 0 mean and a 3 standard deviation was used to simulate the PSF for the generation of the LR-HSI. The spectral response of the IKONOS satellite (a Nikon D700 camera) was used to simulate the SRF for the generation of the HR-MSI.

*2) Real Dataset:* The real dataset is the so-called Houston University dataset. It was released by the 2018 IEEE GRSS Data Fusion Contest and was used as the reference for the evaluation of uEDSSG. Specifically, an LR-HSI and a high-resolution RGB (HR-RGB) image from this multimodal optical remote-sensing dataset were used in the experiment. The LR-HSI acquired from a ITRES CASI-1500 sensor was of size $4172 \times 1202 \times 48$ with a 1-m spatial resolution and the spectral coverage ranging from 0.38 to 1.05 $\mu$m. The HR-RGB acquired from a DiMAC ULTRALIGHT+ sensor was of size $83\,440 \times 24\,040 \times 3$ with a 0.05-m spatial resolution. The observed data of the LR-HSI was downsampled to an area of $64 \times 64 \times 48$, and the corresponding area of the HR-RGB with a $512 \times 512 \times 3$ size was adopted.

*3) Quantitative Metrics:* To evaluate the performance of the fused image from all the aforementioned methods for synthetic datasets, the following six quantitative metrics were adopted: the peak signal-to-noise ratio (PSNR), universal image quality index (UIQI), root mean squared error (RMSE), structured similarity (SSIM), spectral angle mapper (SAM), and erreur relative globale adimensionnelle de synthèse (ERGAS) [49], [50]. The PSNR, RMSE, and UIQI were for evaluating the similarity between the fused image and the reference. The SSIM, SAM, and ERGAS measure the structural consistency between the fused image and the reference. The larger the PSNR, SSIM, and UIQI values, the better the fused result. While the smaller the RMSE, SAM, and ERGAS values, the less the error in the fused image and the less the distortion in the spectral.

Similarly, the fused images resulting from all the methods but based on real datasets were evaluated using the following three quantitative metrics without the use of reference: spectral distortion index Dl, spatial distortion index Ds, and quality with no reference (QNR) [51], [52]. The Dl and Ds were for spectral and spatial distortion, and the QNR was for the quality of the fused results. The smaller the Dl and Ds values, the less the spectral and spatial distortion in the fused results, and the larger the QNR value, the better the fused results.

TABLE I
PSNR, SAM, AND NCE OF THE HR-HSIS FUSED BY UEDSSG AND FOUR SCHEMES

| | Pavia University | | | Indian Pines | | | Washington DC | | |
|---|---|---|---|---|---|---|---|---|---|
| | PSNR | SAM | epoch | PSNR | SAM | epoch | PSNR | SAM | epoch |
| - | 35.01 | 4.02 | 9150 | 33.17 | 5.18 | 6950 | 34.21 | 2.53 | 10100 |
| ReLU | 36.27 | 3.85 | 9250 | 35.91 | 4.04 | 7100 | 35.47 | 2.41 | 10000 |
| PReLU | 37.91 | 3.42 | 11150 | 36.48 | 3.99 | 9000 | 35.82 | 2.38 | 12000 |
| LReLU | 38.16 | 3.07 | 8250 | 36.77 | 3.96 | 5750 | 36.18 | 2.19 | 9750 |

### C. Discussion on Activation Function

To investigate the impact of activation function on uEDSSG, uEDSSG without activation function and uEDSSGs with activation functions were tested in the section. In the test, ReLU, PReLU, and LReLU were selected as the activation function in uEDSSGs due to the properties of unidirectional saturation and low-complexity derivative calculation. In addition, the number of convergence epochs (NCEs) was added as the evaluation indicator of convergence.

Comparisons among PSNR, SAM, and NCE of the fused HR-HSIs based on the three synthetic datasets and the following four schemes were performed: 1) uEDSSG without an activation function; 2) uEDSSG with ReLU; 3) uEDSSG with LReLU; and 4) uEDSSG with PReLU. Table I shows the PSNR, SAM, and NCE results. Compared with all the rest schemes, the PSNR, SAM, and NCE resulting from uEDSSG with LReLU (fourth scheme) are the highest, lowest, and lowest, respectively, indicating that LReLU is the most suitable activation function in the proposed uEDSSG.

### D. Selection for Optimal Number of Endmembers

In general, the accuracy of fusion improves as the number of endmembers increases. However, in the proposed network, the number of endmembers represents the number of abundances and the shared convolution layer. Therefore, the larger the number, the larger the model. To ensure that a smaller model achieves accurate fusion, the experiment for the optimal number of endmembers were performed. In the experiment, the effect of the number of endmembers on the performance of uEDSSG was investigated using the aforementioned three synthetic datasets, and the PSNR and SAM were used to evaluate the fused images. Fig. 8 shows the PSNR and SAM

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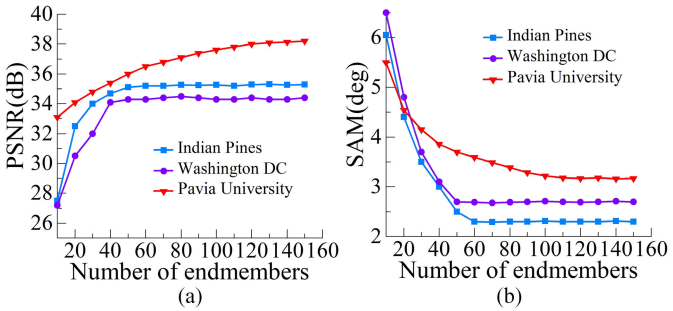



Fig. 8. Variation in (a) PSNR and (b) SAM with the number of endmembers based on three synthetic datasets.

curves of the fused image resulting from various values for the number of endmembers. We can find that the PSNR and SAM results from all the three datasets significantly improve with the increase in the number of endmembers, especially in the range from 10 to 50. The fusion accuracy from the Indian Pines (blue) and Washington DC (purple) data, begins to converge when the number of endmembers is about 70. The results from the Pavia University data (red) begin to converge when the number of endmembers is about 100. Compared with the results of the other two datasets, the result of the Pavia University dataset (red) shows a slower convergence. Based on these results, the value of 100 was regarded the optimal number of endmembers, thus this value was adopted in the experiment.

### *E. Ablation Test*

To investigate the necessity of each module and the joint loss function used in uEDSSG, this section presents the results of ablation tests for each module and the loss function. For verifying the improvement in HSI–MSI fusion made by the use of multihierarchical encoders with spatial and spectral guidance, and deep coupling of PSF/SRF and edge of HSIs, comparisons among the fused HR-HSIs based on the three synthetic datasets and the following four schemes were performed: 1) uEDSSG; 2) uEDSSG without multihierarchical encoders ($\not\subset$ MHE); 3) uEDSSG without spatial and spectral guidance ($\not\subset$ SG); and 4) uEDSSG without deep coupling of PSF/SRF and edge of HSIs ($\not\subset$ dc). In particular, uEDSSG without MHE (i.e., the multihierarchical encoders were removed and replaced with the encoders that were composed of 2-D convolution layers and LReLU layers alternately). Table II shows the PSNR and SAM results. Compared with all the rest schemes, the PSNR and SAM resulting from uEDSSG (the first scheme) are the highest and lowest, respectively. These results imply that the usage of multihierarchical encoders with spatial and spectral guidance, and deep coupling of PSF/SRF and edge of HSIs can constrain spatial and spectral details more effectively.

For evaluating the improvement in the HSI–MSI fusion made by each term in the joint loss function expressed by (20), experiments using the same three synthetic datasets and the following five schemes were carried out: 1) uEDSSG [i.e., (20)]; 2) uEDSSG without Lspaspe ($\not\subset \mathcal{L}_{\text{spaspe}}$); 3) uEDSSG without $\mathcal{L}_{\nabla}$ ( $\not\subset \mathcal{L}_{\nabla}$); 4) uEDSSG without Lsum2one ($\not\subset \mathcal{L}_{\text{sum2one}}$); and 5) uEDSSG without Lsparse ($\not\subset \mathcal{L}_{\text{sparse}}$). Their resultant HR-HSIs are shown in Fig. 9 in the form of a histogram. Compared with all the rest schemes, PSNR obtained by uEDSSG is the highest for all the three datasets. This suggests the joint loss function expressed by (20) plays an important role in improving the fusion performance.

TABLE II
PSNR AND SAM OF THE HR-HSIS FUSED BY UEDSSG AND FOUR SCHEMES

| | PQIs | uEDSSG | $\not\subset$MHE | $\not\subset$SG | $\not\subset$DC |
|---|---|---|---|---|---|
| Pavia University | PSNR | 38.71 | 37.18 | 37.74 | 37.49 |
| | SAM | 3.01 | 3.27 | 3.19 | 3.23 |
| Indian Pines | PSNR | 36.16 | 35.24 | 35.46 | 35.61 |
| | SAM | 2.11 | 2.43 | 2.28 | 2.21 |
| Washington DC | PSNR | 36.01 | 35.13 | 35.32 | 35.57 |
| | SAM | 2.35 | 2.79 | 2.61 | 2.54 |

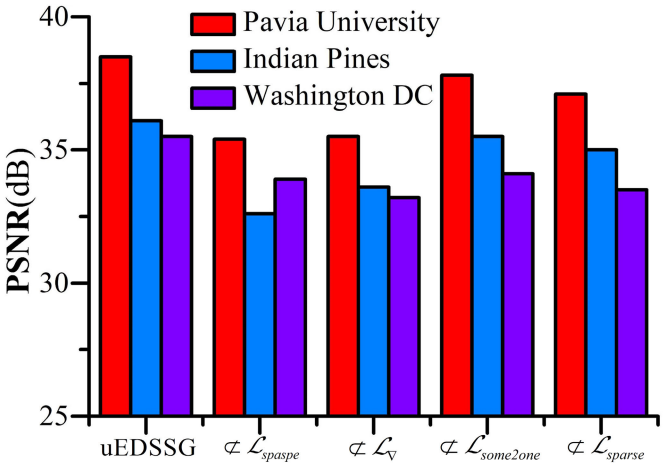



Fig. 9. PSNR of the HR-HSIs fused by five schemes: the first one (uEDSSG) contains all terms in the joint loss function, and the rest four were formed by excluding one term in the joint loss function.

### *F. Estimation of PSF and SRF*

In the selected three synthetic datasets, the aforementioned LR-HSI and HR-MSI were generated by simulations using a PSF and an SRF, respectively. To investigate the PSF and SRF estimated by uEDSSG, uEDSSG and each synthetic dataset were used to learn the PSF and SRF, and the PSF and SRF results are shown in Figs. 10 and 11, respectively.

In Fig. 10, the first column shows the true PSFs generated using the Gaussian kernel with the standard deviation values of 1 (upper), 2 (middle), and 3 (bottom); the other three columns are the corresponding PSFs estimated by uEDSSG. The kernel sizes of the upper, middle, and bottom rows were set to 8, 16, and 16, respectively. From all the rows, it can be seen that the estimated PSFs are similar to the true ones, meaning that uEDSSG is suitable for estimating arbitrary PSFs based on different datasets.

The SRF estimated from the datasets of the Pavia University, Indian Pines, and Washington DC datasets are shown in the

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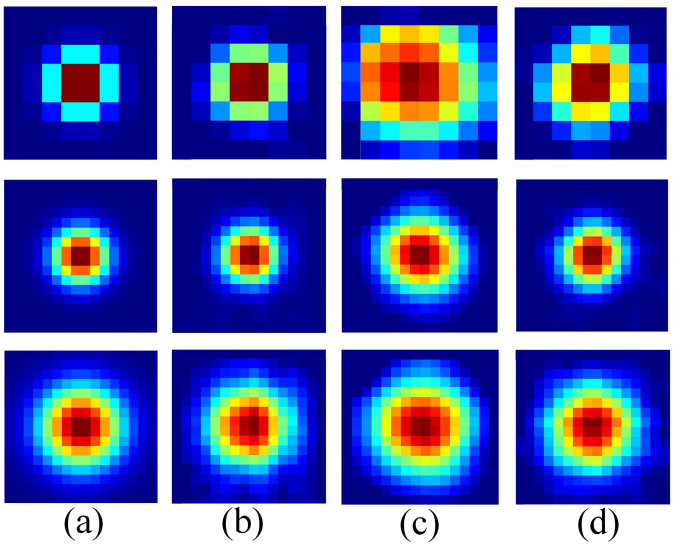



Fig. 10. (a) (Left column) True PSFs generated by the Gaussian kernel with the standard deviations of (upper row) 1, (middle row) 2, and (bottom row) 3, and the corresponding PSF estimated by uEDSSG based on the datasets of (b) Pavia University, (c) Indian Pines, and (d) Washington DC.

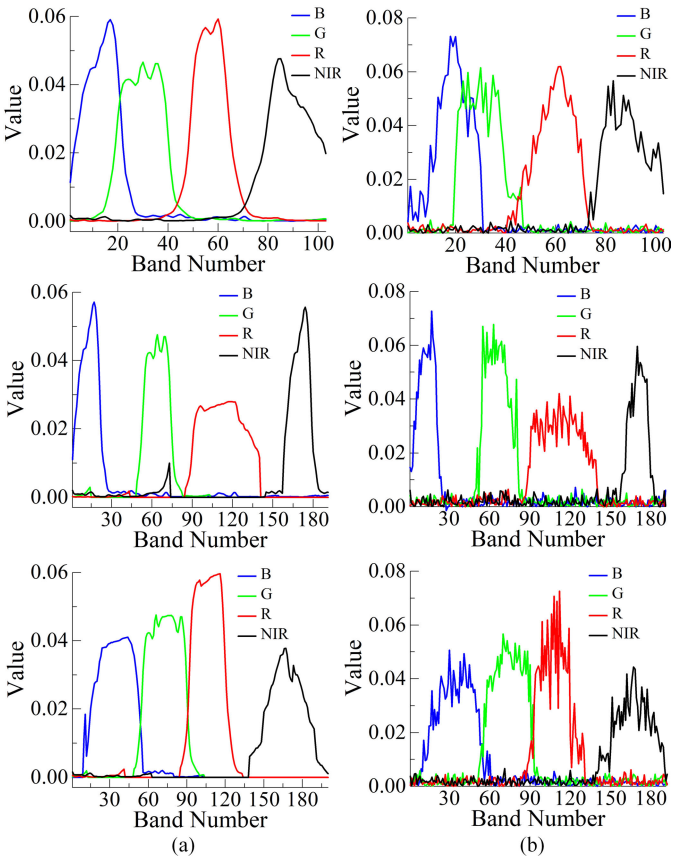



Fig. 11. (a) (Left column) True SRF generated by spectral response of a Nikon D700 camera and (b) SRF estimated (right column) by uEDSSG from the datasets of (upper) Pavia University, (middle) Indian Pines, and (bottom) Washington DC.

upper, middle, and bottom rows in Fig. 11(b), respectively. The three figures shown in Fig. 11(a) (the left column) are the true PSFs, which were generated by the spectral response of a Nikon D700 camera, and the right column shows the estimated results. Comparing the two figures in the same row, we can see that although the curves of the estimated SRF are not as smooth as the true ones, both are slightly different in the trend of their variation with band number. This result indicates that the proposed uEDSSG can roughly reflect the SRFs based on different datasets.

### *G. Results From Different Synthetic Datasets*

The performance of uEDSSG based on each of the three synthetic datasets was compared with that of the following seven methods named: Hysure [10], OTD [21], LTTR [37], NLSTF [30], uSDN [38], MHFnet [40], and Hyconet [47]. Among these methods, Hysure, MHFnet, Hyconet, and uEDSSG do not need an SRF and an PSF. For an appropriate comparison, the SRFs and PSFs estimated by HySure were used to compare with those methods that require an SRF and PSF as input. To account for ubiquitous noise or error, Gaussian noise (SNR = 20/30 dB) was added to the LR-HSI and the HR-MSI of the three synthetic datasets.

Fig. 12 shows the images fused from the above-mentioned eight methods (the first eight columns) and the reference image (the last column) which was obtained also from the Pavia University dataset. It is noted that the first/third row is resulted from the Pavia University dataset with Gaussian noise (SNR = 20/30 dB) in the form of the 30th band gray and the second/fourth row is its corresponding error map which is the difference of the fused image from its reference image. The fusion results of HySure and uEDSSG on the dataset with Gaussian noise (SNR = 30 dB) are roughly similar to those on the dataset with SNR = 20 dB, while the fusion results of the remaining six methods on the dataset with Gaussian noise (SNR = 30 dB) are poorer. Comparing all the fused images in the first/third rows, we can find that the objects shown in Fig. 12(d) (i.e., the result of NLSTF) is spatially ambiguous, meaning the fused result of NLSTF Fig. 12(d) misses the spatial details. Comparing all error maps in the second/fourth row, Hyconet [Fig. 12(g)] and uEDSSG [Fig. 12(h)] are insignificant, meaning that their fused images are the ones closest to the reference image [Fig. 12(i)].

Similar to Fig. 12, Fig. 13 shows the images fused from the same eight methods but based on the Washington DC dataset in form of the 50th band gray. We can observe that the fusion results of MHFnet and uEDSSG are almost unaffected when the SNR of Gaussian noise increases from 20 to 30 dB. The error shown in NLSTF [Fig. 13(d)] is the most significant (the worst one), while uEDSSG [Fig. 13(h)] is the best. Fig. 14 shows the fused images from the Indian Pines dataset in form of the 100th band gray. One can see that HySure [Fig. 14(a)] and NLSTF [Fig. 14(d)] are the worst, Hyconet [Fig. 14(g)] and uEDSSG [Fig. 14(h)] are the best ones, and uEDSSG can resist the interference of noise.

The above figures are for visual comparisons. For quantitative comparisons, the variation in the PSNR value with the band in the HR-HSIs fused from each method

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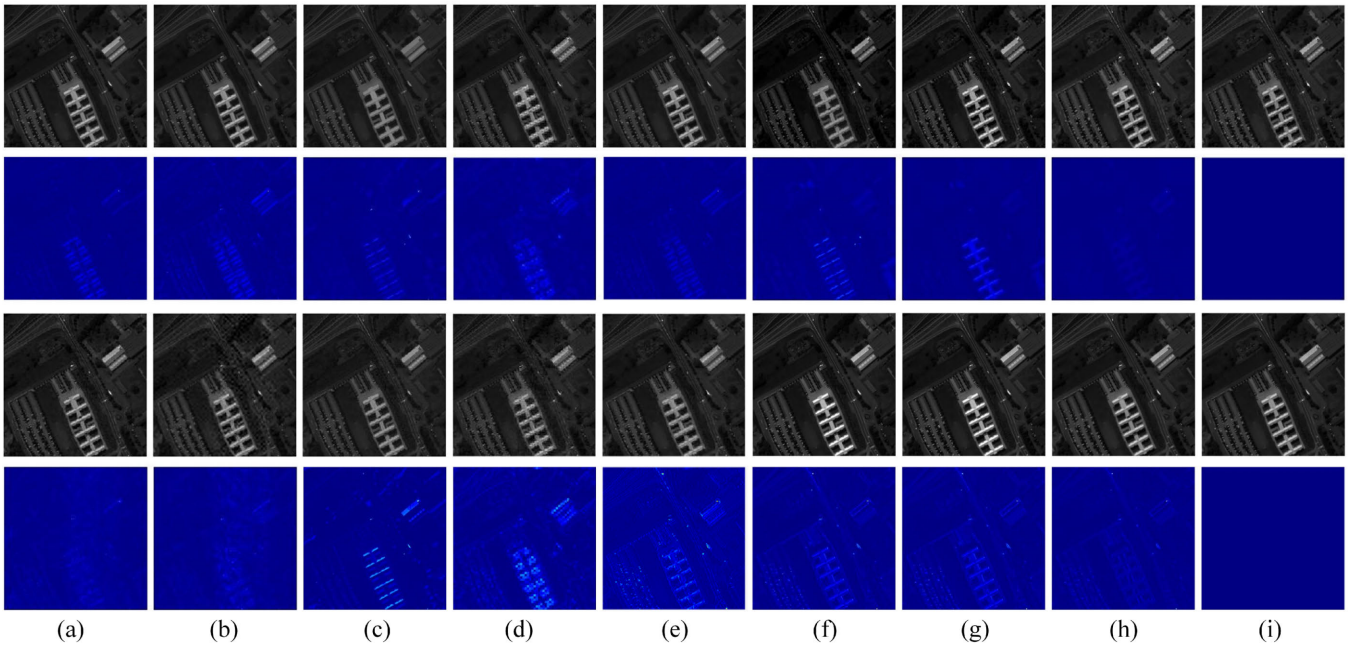



Fig. 12. (First/third row) Fused images and (second/fourth row) error maps at band 30 resulting from the Pavia University dataset with Gaussian noise (SNR = 20/30 dB) and the methods of (a) HySure, (b) OTD, (c) LTTR, (d) NLSTF, (e) uSDN, (f) MHFnet, (g) Hyconet, and (h) uEDSSG. The last column (i) is the reference image. The error map was obtained from the difference between the fused and reference images.

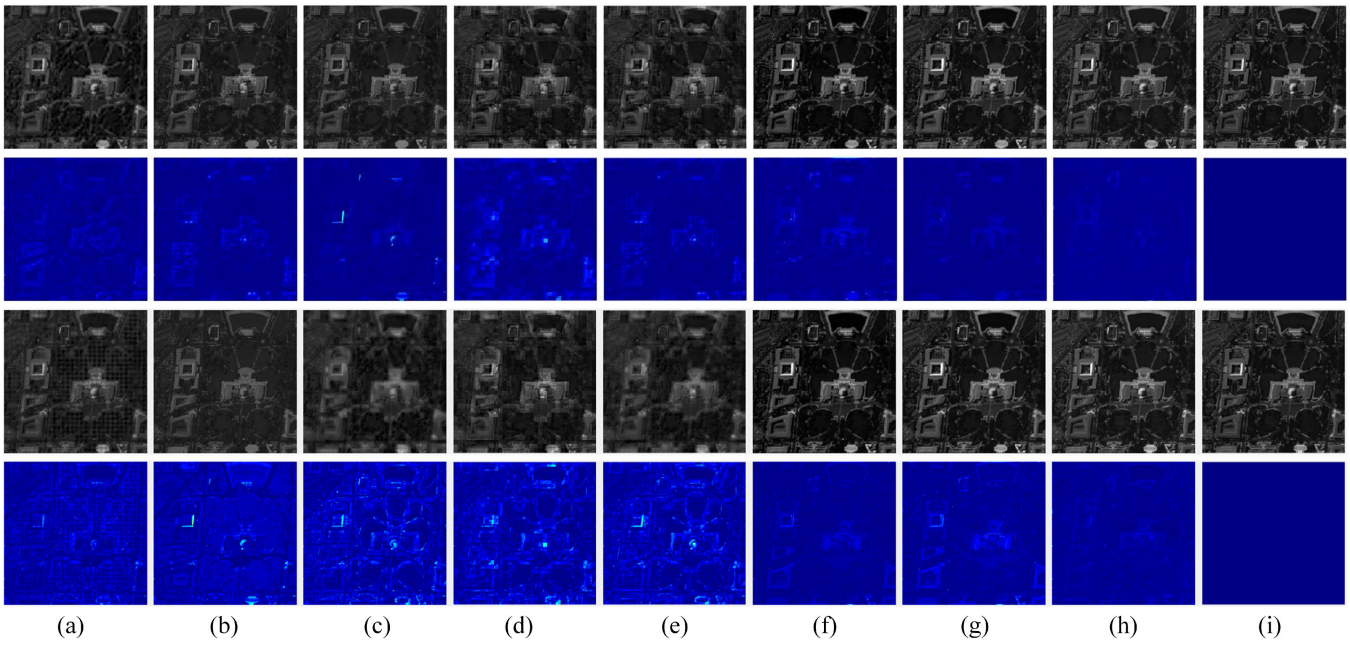



Fig. 13. (First/third row) Fused images and (second/fourth row) error maps at band 50 resulting from the Washington DC dataset with Gaussian noise (SNR = 20/30 dB) and the methods of (a) HySure, (b) OTD, (c) LTTR, (d) NLSTF, (e) uSDN, (f) MHFnet, (g) Hyconet, and (h) uEDSSG. The last column (i) is the reference image.

and each dataset is shown in Fig. 15, where Fig. 15(a)–(c) shows the results from the Pavia University, Washington DC, and Indian Pines datasets, respectively. It is noted that the upper and bottom rows are resulted from the three datasets with Gaussian noise (SNR = 20/30 dB), respectively. All results indicate that, generally, the PSNR of uEDSSG (red) is almost the largest in each figure on most of the bands. It can be concluded that the proposed uEDSSG method outperforms the other methods in maintaining spatial information.

For overall quality evaluation, all the PSNR, RMSE, UIQI, SSIM, SAM, and ERGAS values of the HR-HSIs obtained from all the selected methods and the three synthetic datasets with Gaussian noise (SNR = 20/30 dB) as shown in Table III. The results suggest uEDSSG is the best performer in terms of all the above six PQIs.

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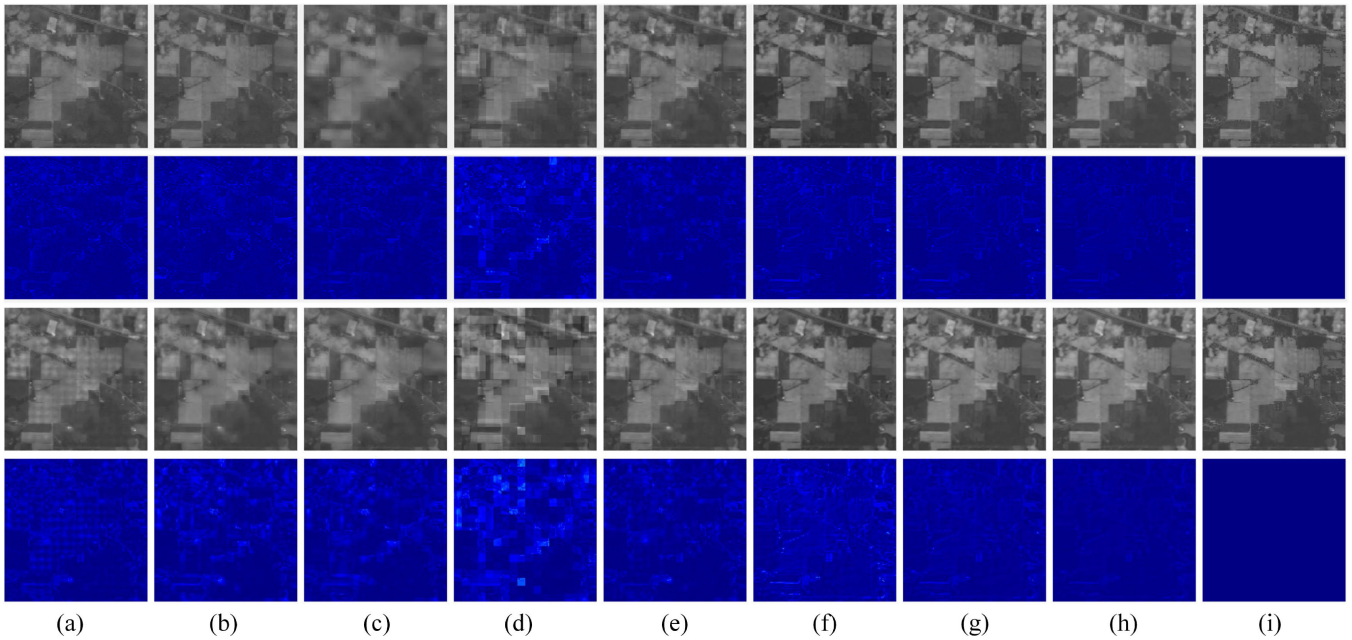



Fig. 14. (First/third row) Fused images and (second/fourth row) error maps at band 100 resulting from the Indian Pines dataset with Gaussian noise (SNR = 20/30 dB) and the methods of (a) HySure, (b) OTD, (c) LTTR, (d) NLSTF, (e) uSDN, (f) MHFnet, (g) Hyconet, and (h) uEDSSG. The last column (i) is the reference image.

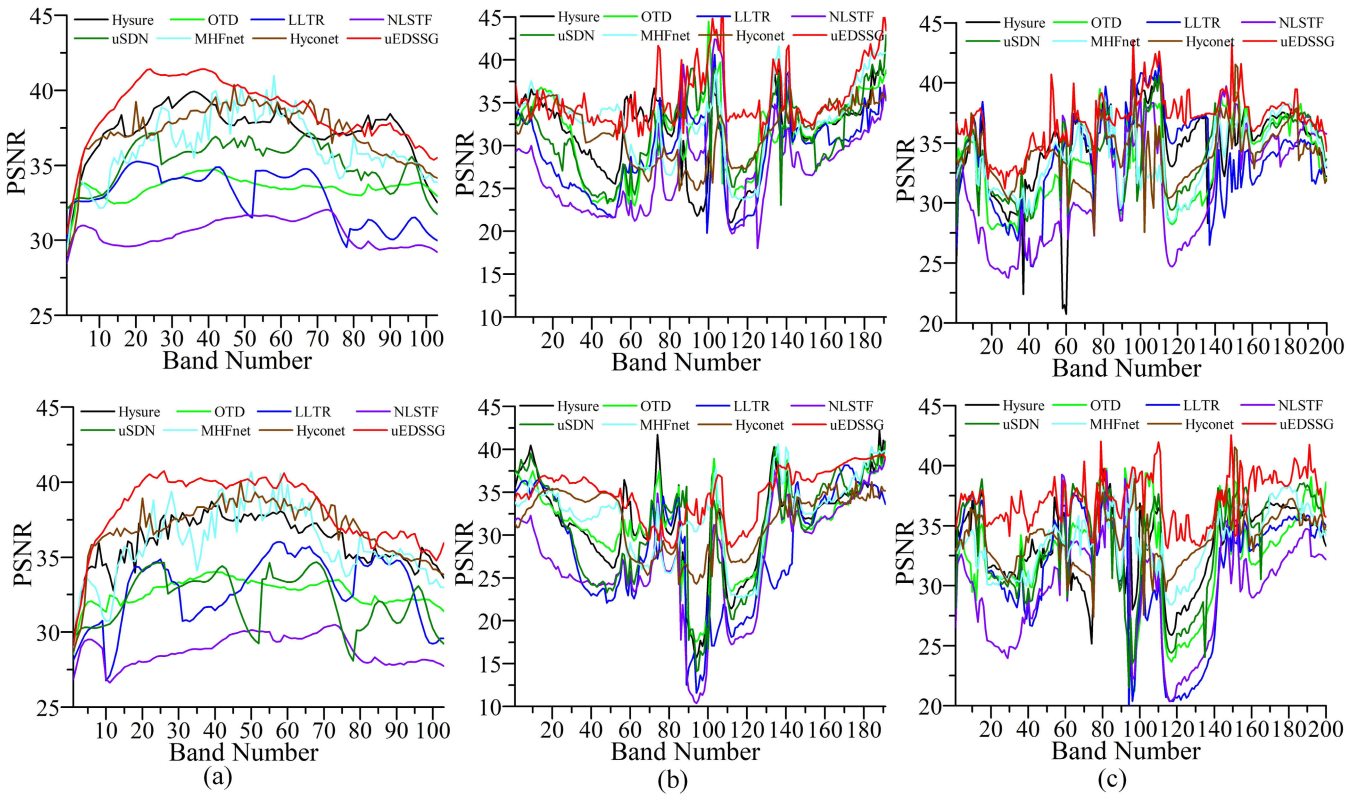



Fig. 15. Variation in PSNR with spectral band resulting from the selected eight methods and the datasets of (a) Pavia University, (b) Washington DC, and (c) Indian Pines with Gaussian noise SNR = 20 (upper row) and 30 dB (bottom row).

### *H. Results From the Real Dataset*

For practice, the performance of uEDSSG using the real dataset: the Houston University dataset was compared with that of seven existing methods, and results are shown in Fig. 16. In each figure, the large red box is a zoom-in of the small red box in the same figure merely for a clearer observation. Fig. 16(a) shows the HR-RGB image. Fig. 16(b) is the RGB image of the LR-HSI; the remaining figures show the RGB images of the HR-HSIs resulting from the eight

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TABLE III

QUANTITATIVE METRICS OF THE HR-HSIS FUSED BY EIGHT METHODS AND THE DATASETS OF (TOP) PAVIA UNIVERSITY, (MIDDLE) WASHINGTON DC, AND (BOTTOM) INDIAN PINES WITH GAUSSIAN NOISE (SNR = 20/30 DB)

| HSIs | SNR | PQIs | Hysure | OTD | LLTR | NLSTF | uSDN | MHFnet | Hyconet | uEDSSG |
|---|---|---|---|---|---|---|---|---|---|---|
|  **Pavia University** | 20 | PSNR | 36.15 | 35.34 | 33.8 | 32.16 | 36.74 | 37.16 | 37.98 | 38.19 |
| | | RMSE | 4.05 | 5.12 | 7.09 | 6.71 | 3.24 | 3.99 | 1.93 | 2.48 |
| | | SSIM | 0.96 | 0.94 | 0.84 | 0.89 | 0.96 | 0.96 | 0.96 | 0.97 |
| | | ERGAS | 2.41 | 3 | 5.57 | 5.18 | 2.73 | 1.64 | 1.07 | 0.49 |
| | | UIQI | 0.97 | 0.94 | 0.89 | 0.9 | 0.97 | 0.98 | 0.98 | 0.99 |
| | | SAM | 4.01 | 3.9 | 5.43 | 5.45 | 3.78 | 3.61 | 3.3 | 3.06 |
| | 30 | PSNR | 35.92 | 33.87 | 31.95 | 31.7 | 35.23 | 36.31 | 36.25 | 38.18 |
| | | RMSE | 4.08 | 6.52 | 6.92 | 7.71 | 4.18 | 4.05 | 3.05 | 2.58 |
| | | SSIM | 0.96 | 0.94 | 0.81 | 0.87 | 0.96 | 0.96 | 0.96 | 0.97 |
| | | ERGAS | 2.43 | 4.48 | 7.29 | 6.13 | 1.64 | 2.38 | 2.39 | 0.5 |
| | | UIQI | 0.97 | 0.93 | 0.87 | 0.89 | 0.97 | 0.97 | 0.97 | 0.99 |
| | | SAM | 4.02 | 4.34 | 8.49 | 6.04 | 4.12 | 3.95 | 3.97 | 3.07 |
|  **Washington DC** | 20 | PSNR | 33.259 | 33.19 | 32.44 | 30.59 | 33.44 | 34.52 | 35.25 | 36.77 |
| | | RMSE | 4.56 | 5.05 | 7.01 | 9.13 | 4.05 | 3.95 | 1.84 | 1.58 |
| | | SSIM | 0.95 | 0.94 | 0.9 | 0.89 | 0.97 | 0.96 | 0.98 | 0.99 |
| | | ERGAS | 2.08 | 2.25 | 3.57 | 5.1 | 1.8 | 1.73 | 1. 54 | 1.44 |
| | | UIQI | 0.87 | 0.86 | 0.78 | 0.84 | 0.91 | 0.96 | 0.97 | 0.98 |
| | | SAM | 5.28 | 5.36 | 6.9 | 8.77 | 4.51 | 4.23 | 4.09 | 3.96 |
| | 30 | PSNR | 32.97 | 31.84 | 31.3 | 29.11 | 31.44 | 34.43 | 34.11 | 36.75 |
| | | RMSE | 4.93 | 8.85 | 7.41 | 9.65 | 8.24 | 3.99 | 4.05 | 1.58 |
| | | SSIM | 0.95 | 0.9 | 0.88 | 0.84 | 0.89 | 0.96 | 0.96 | 0.99 |
| | | ERGAS | 3.12 | 4.21 | 4.89 | 6.03 | 4.63 | 1.82 | 1. 91 | 1.44 |
| | | UIQI | 0.83 | 0.77 | 0.69 | 0.7 | 0.81 | 0.96 | 0.95 | 0.98 |
| | | SAM | 5.53 | 7.59 | 8.98 | 9.95 | 8.51 | 4.31 | 4.64 | 3.99 |
|  **Indian_pines** | 20 | PSNR | 33.67 | 33.44 | 31.97 | 32.63 | 33.64 | 34.23 | 35.09 | 36.17 |
| | | RMSE | 3.49 | 4.13 | 4.59 | 4.07 | 3.51 | 3.12 | 2.09 | 1.94 |
| | | SSIM | 0.95 | 0.94 | 0.92 | 0.93 | 0.95 | 0.95 | 0.95 | 0. 96 |
| | | ERGAS | 1.5 | 1.57 | 2.99 | 2.69 | 1.55 | 1.46 | 1.43 | 1.38 |
| | | UIQI | 0.84 | 0.81 | 0.8 | 0.77 | 0.84 | 0.89 | 0.92 | 0.95 |
| | | SAM | 2.41 | 2.42 | 2.86 | 2.75 | 2.39 | 2.37 | 2.27 | 2.2 |
| | 30 | PSNR | 32.73 | 31.31 | 30.9 | 28.3 | 32.81 | 33.56 | 33.87 | 36.15 |
| | | RMSE | 4.03 | 7.46 | 5.42 | 9.64 | 4.04 | 3.43 | 3.59 | 1.95 |
| | | SSIM | 0.93 | 0.92 | 0.89 | 0.81 | 0.93 | 0.94 | 0.94 | 0. 96 |
| | | ERGAS | 1.58 | 3.23 | 3.75 | 4.78 | 1.68 | 2.18 | 1.63 | 1.38 |
| | | UIQI | 0.79 | 0.78 | 0.74 | 0.65 | 0.8 | 0.84 | 0.85 | 0.95 |
| | | SAM | 2.88 | 3.83 | 3.18 | 4.24 | 2.67 | 2.43 | 2.39 | 2.21 |

TABLE IV

QUANTITATIVE METRICS OF THE HR-HSIS FUSED BY EIGHT METHODS FOR THE HOUSTON UNIVERSITY DATASET

| PQIs | Hysure | OTD | LLTR | NLSTF | uSDN | MHFnet | Hyconet | uEDSSG |
|---|---|---|---|---|---|---|---|---|
| Dl | 0.0683 | 0.0575 | 0.0794 | 0.0773 | 0.0621 | 0.0891 | 0.0803 | 0.0518 |
| Ds | 0.1938 | 0.1814 | 0.1362 | 0.2385 | 0.1041 | 0.0956 | 0.0892 | 0.0734 |
| QNR | 0.7511 | 0.7715 | 0.81 | 0.6807 | 0.8062 | 0.8097 | 0.8165 | 0.8322 |

TABLE V

RUNNING TIME OF THE HR-HSIS FUSED BY EIGHT METHODS AND FOUR DATASETS

| HSIs | Hysure | OTD | LLTR | NLSTF | uSDN | MHFnet | Hyconet | uEDSSG |
|---|---|---|---|---|---|---|---|---|
| Pavia University | 213.4 | 172.4 | 434.1 | 290.4 | 678.3 | 953.4 | 753.5 | 974.6 |
| Washington DC | 215.8 | 176.3 | 435.5 | 278.7 | 634.1 | 935.7 | 759.7 | 963.2 |
| Indian_pines | 193.1 | 154.7 | 426.3 | 284.1 | 647.2 | 931.5 | 769.4 | 947.5 |
| Houston University | 209.7 | 135.8 | 384.7 | 246.3 | 385.4 | 792.3 | 614.1 | 813.4 |

methods. Note that the colorful spatial details shown in the area of the small red boxes are convenient for comparison between spatial and spectral details. Comparing the results among all the large red boxes, we can see that Fig. 16(c) HySure, Fig. 16(f) NLSTF, and Fig. 16(j) uEDSSG present good results in terms of color and brightness in the spectral aspect, while the result of the proposed uEDSSG method is much closer to the HR-RGB image than all the other methods in the spatial aspect.

Table IV shows the three measures selected for the quality assessment of the fusion results from the eight methods shown in Fig. 16 and the Houston University dataset. It can be seen that uEDSSG leads to the lowest Dl, lowest Ds, and highest QNR, meaning that uEDSSG is the most effective method for fused HR-HSIs.

### *I. Computational Efficiency*

The computational efficiencies of all the aforementioned eight methods and four datasets were also compared, as shown in Table V. It is worth mentioning that the first four methods including Hysure, OTD, LLTR, and NLSTF were executed using MATLAB, and the remaining three network methods, uSDN, MHFnet, Hyconet, and uEDSSG, were implemented

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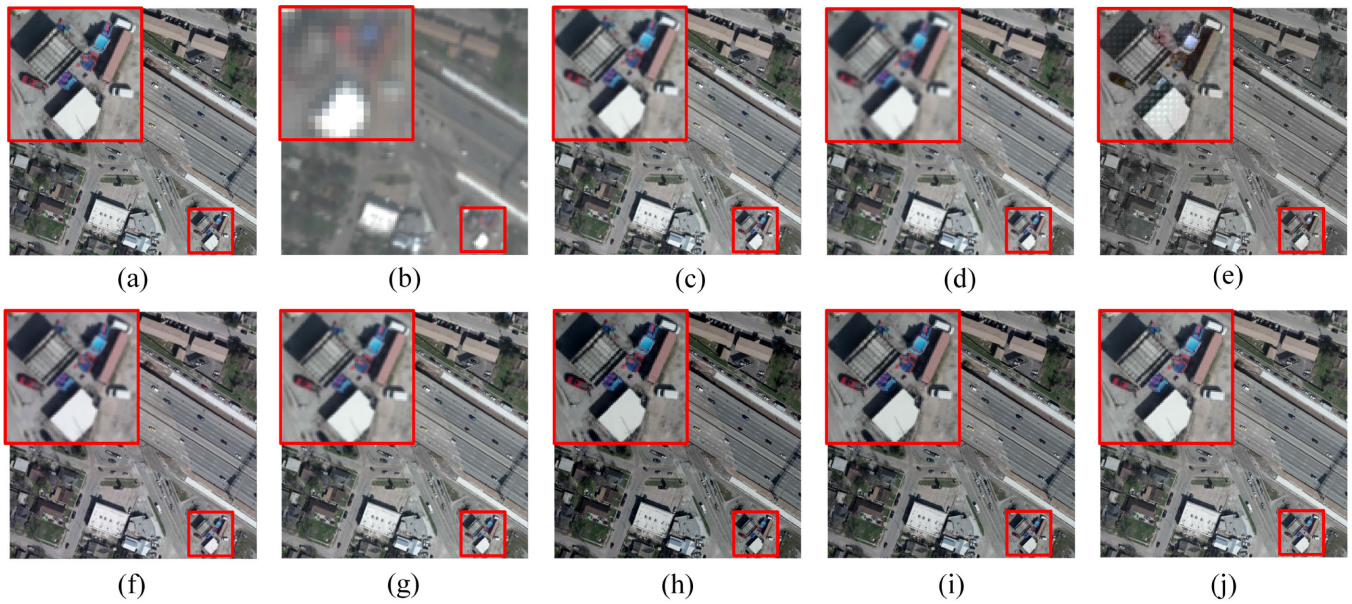



Fig. 16. (a) HR-RGB image and (b) RGB image of LR-HSI, and RGB images of the results fused from the real dataset with the methods of (c) HySure, (d) OTD, (e) LTTR, (f) NLSTF, (g) uSDN, (h) MHFnet, (i) Hyconet, and (j) uEDSSG.

by the PyTorch framework. From Table V, it can be seen that the running times of the first five methods are shorter than the training times of the three network methods. The main time burden of uEDSSG is from spatial and spectral guidance.

## V. Conclusion

In this study, to maintain the correlation between spectral and spatial domains in an HR-HSI, an unsupervised encoder–decoder network under spatial and spectral guidance for HSI–MSI fusion (uEDSSG) was proposed. Multihierarchical encoders under spatial and spectral guidance injected spatial information from an HR-MSI and spectral information from an LR-HSI into the extraction of the abundances of the LR-HSI and HR-MSI, respectively. The deep coupling of PSF/SRF and edge of HSIs and a spatial–spectral constraint were constructed for maintaining spatial and spectral details. Experimental results based on three synthetic datasets and one real dataset showed that the uEDSSG method can extract spatial information from the HR-MSI and also restore spectral information in the LR-HSI. The proposed uEDSSG can effectively improve the result of HSI–MSI fusion.

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